

Noticing Where the Forest Folds: Exploring (Mushroom) Foraging as a More-Than-Human Design Method

Fredrik Bernerstedt
Department of Informatics and Media
Uppsala University
Uppsala, Sweden
fredrik.bernerstedt.2156@student.uu.se

Jingpeng Feng
Department of Informatics and Media
Uppsala University
Uppsala, Sweden
jingpeng.feng.9029@student.uu.se

YuHua Huang
Department of Informatics and Media
Uppsala University
Uppsala, Sweden
yu-hua.huang.3888@student.uu.se

Rahul Reddy Tikkavarapu
Department of Informatics and Media
Uppsala University
Uppsala, Sweden
rahul-
reddy.tikkavarapu.6596@student.uu.se

Jon Back
Department of Informatics and Media
Uppsala University
Uppsala, Sweden
jon.back@im.uu.se

Abstract

The ongoing post-anthropocentric turn within HCI is fueled by the drive to revisit the relationship between human and nature as a result of urbanization, resource exploitation and lifestyle changes. However, within more-than-human design, we observe that most existing practices tend to build indirect experience of nature engagement rather than fostering direct, embodied encounters with the natural environment. In this work, we ask *how foraging activities, such as mushroom picking, can be re-framed as a noticing practice within more-than-human design*, thereby engaging directly with nature through curiosity and noticing. We present the results from two in-the-wild design workshops located in nature, and discuss how they changed the outcome to be more embedded with the more-than-human perspective. Finally, two design principles are presented, intended to further more-than-human design by encouraging *movement through space*, and *fluid scale-shifting* between macro- and micro-level. Thereby gaining a comprehensive understanding of the environment we design for.

CCS Concepts

• Human-centered computing → HCI design and evaluation methods.

Keywords

nature, design method, more-than-human design, noticing, in-the-wild

ACM Reference Format:

Fredrik Bernerstedt, Jingpeng Feng, YuHua Huang, Rahul Reddy Tikkavarapu, and Jon Back. 2026. Noticing Where the Forest Folds: Exploring (Mushroom) Foraging as a More-Than-Human Design Method. In *Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems (CHI EA '26)*, April 13–17, 2026, Barcelona, Spain. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3772363.3798654>

1 Introduction

While it has been shown that closeness to nature helps people recover from various deficits such as attentive loss and stress, and return to a somewhat baseline state of well-being, the modern day societal tendencies of urbanization, resource exploitation and lifestyle changes have diminished our possibilities to have contact with nature [7]. Within this context, HCI has explored a wide range of technologies to mediate the relationship between humans and the nature worlds [29]. By reviewing past HCI works on human-nature relationships, Webber et al. [29] identified an ongoing diversification of perspectives, including increasing attention to more-than-human interactions, that aim to reframe human-nature relationships beyond anthropocentric viewpoints.

The evident post-anthropocentric turn in HCI is fueled by global concerns with environmental sustainability as a result of human activities [15, 19]. We have seen scholars investigating ways of decentering humans, leaving the spotlight to be shared with non-human entities through the practice and development of design methodologies that do not solely focus on the objects that we produce, but also on these processes themselves [15]. Echoing this paradigm shift, we made our first attempt through a design workshop focused on designing for plants. The workshop served as a medium for introducing the concept of a post-anthropocentric perspective, aiming to influence participants through a series of design scaffolds. However, within the two-hour session, we recognized that our influence on participants was limited. The final prototypes they built to a large extent remained a highly human-centered perspective.

More-than-human researchers have introduced various methods and design practices aimed at cultivating a post-anthropocentric perspective. One of these approaches is the *Art of Noticing*, described by anthropologist Anna Tsing [25] as a mindset where you look around you rather than looking ahead, throwing aside classical narratives of progress and embracing curiosity, not least about multispecies lifeworlds, but how they are impacted by economical, ecological, political and cultural practices. From an HCI standpoint, we may view these multispecies lifeworlds and as entangled with our practices, in addition to technology, thus making way for a



This work is licensed under a Creative Commons Attribution 4.0 International License. *CHI EA '26, Barcelona, Spain*

© 2026 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-2281-3/26/04

<https://doi.org/10.1145/3772363.3798654>

responsive and responsible school of design in which every(thing) can thrive [24].

Even within more-than-human design (including our own work leading up to this), we observe that most existing practices tend to build *indirect experience* [29] of nature engagement, rather than fostering direct, embodied encounters with the natural environment. Therefore, in this work, we ask *how the direct nature experience of foraging activities can inform more-than-human design and research, and more specifically, the method of the Art of Noticing*. To explore this question, we performed two design workshops and two evaluation sessions in the forest, inspired by the culturally embedded practice of mushroom foraging in Sweden, and from these propose design principles to inform future noticing design for direct nature engagement in more-than-human design.

2 Related Work

In the HCI domain, researchers have long explored how technology can be used to foster connections between humans and the natural environment. In this section, we focus specifically on related work concerning nature, Swedish-style foraging practice, and more-than-human noticing.

2.1 HCI and Nature

Studies are often grounded in the idea that nature serves as an embodied medium that supports human well-being [31] and learning [26]. Within this area, digital technologies have been designed to bridge experiences between human and nature in restorative [5, 14, 20], playful [9, 18], and educational [36] ways. However, these studies take a strongly human-centered stance, viewing nature primarily as a resource for human benefit. This tendency has been identified by Webber et al. [29], who further advocate for this area toward design that acknowledges human-nature *entanglement*. Building on this *entanglement* [6] paradigm, recent HCI researchers have begun to rethink the human-nature relationship by decentering humans, exemplified by engagement with non-human entities including plants [8, 30, 35], microorganisms [34], fungi [11], creek [32], insect [27] and animals [2]. However, in many of these studies, the non-human entities are not situated within the natural environments, but rather in *backyards* [28] or similar controlled areas around the researchers, which are artificially created through technological interventions [8, 30, 34, 35]. Meanwhile, some practices that actually take place in the natural settings tend to lose touch with the everyday and long-term practices that the former can support [11, 32]. To address this gap, our practice aims to combine the advantages of both approaches, exploring potential ways to carry out more-than-human practices in natural environments over the possible long term.

2.2 Swedish-Style Foraging Practice

The foraging culture in Sweden is grounded in the concept of *Allemansrätten* (every man's right), which refers to "the freedom for everyone to roam, and to some extent stay on any landowner's property regardless of their will, to make use of the water in lakes and streams for boating, bathing, laundry etc. and to harvest some wild growing products, such as common berries, flowers, herbs and mushrooms [22]." It is a *customary law* originating in the 1930s

that constitutes a recreational way of escaping urban life [23], and it has since become embedded in the cultural fabric of Swedish society, symbolizing national identity and a sense of *Swedishness* [21]. Existing technologies support foraging through collaborative mapping tools [1, 33] and plant identification application [33]. However, there were some work points toward the more-than-human orientation: Liu et al. [11] developed an embodied technological aid that augments human sensibility, transforming the searching process of foraging into finding suitable habitats for fungi cultivation. Núñez Pacheco and Poikolainen Rosén [16] reflect on how bodily encounters with mushrooms in the wild can cultivate an affective connection to non-human entities. Building on these perspective, this study perceives Swedish-style foraging as a vehicle for informing more-than-human design and future design within nature.

2.3 More-than-Human Noticing

The *Art of Noticing* is one of many methods in the emerging field of MthD (more-than-human design) that has been adopted by the HCI community for the purpose of decentering humans, and account for multispecies and their agency [3, 15]. In related work, it is seen combined with ethnography [12, 25] and auto-ethnography [13, 28]. Often as a means to understand and reflect on different facets of nature through its relation to different practices like agriculture [12], gardening [28], mushroom picking [25] and bird watching [2], or to connect with the environment and forces such as the sun [13] or weather [17]. Within MthD, these efforts are tied to unveiling design opportunities, such as technological interventions in nature and for supporting noticing for researchers and designers so that they may do work towards a sustainable world [3, 13]. Continuing on the trend to understand nature through practices, we see the relation between nature and the practice of foraging mushrooms as a means of noticing in the setting of the Swedish forest. The aim being to promote practices in HCI that supports a holistic perspective of worldbuilding that includes multispecies.

3 Method

In this work, two design sessions and two evaluation sessions were conducted in forest environments in Sweden. During the in-the-wild design sessions, researchers began with a period of hiking in the forest while taking notes on their observations. Each session ended with a sit-down discussion, during which design ideas were generated based on the collected notes and insights. The sessions resulted in a set of design themes, of which four were chosen for continued development, and combined into a design prototype. After design, the prototype was brought back to the forest and evaluated twice, using a semi-functional 'Wizard of Oz'-type prototype [10]. In the first session, the design was internally tested by the researchers, and changes were made in accordance with evaluation results. In the second session, 2 external users were invited based on their prior experience with foraging, both participants had previously tried mushroom picking. Before the testing session, they were provided with information and asked for oral consent, acknowledging their voluntary participation and right to withdraw at any time. One researcher was running the Wizard of Oz-prototype, and two acted as observers, taking notes on participants' behaviors

throughout the process. After the session, an in-depth interview was conducted to capture participants' experiences and reflections. An inductive thematic analysis [4] was performed on observation notes and interviews, where authors from different backgrounds allowed for diverse cultural perspectives on the practice of foraging and interpretation of the results. The analysis resulted in three main themes, reported on below.

4 Design

During the first in-the-wild session, four themes were selected: the art of noticing, embodied attunement, storytelling, and observation aid. Based in these themes, a design inspired by mushroom picking practices was developed, transforming conventional mushroom picking into a design probe for more-than-human noticing practice. Throughout the experience, a deeper relationship between participants, foraging targets, and the surrounding forest are fostered. In particular, the design probe delivers short prompts to support users in discovering their foraging targets and the surrounding. The probe is designed for use in natural environments, particularly forest settings, and in its current version the prompt materials are developed specifically around chanterelle mushrooms.

At the beginning of the experience, users received an initial prompt that framed the interaction as an encounter with *Nature*, a chat-bot character that the user interacted with throughout the experience. *Nature* communicates in first person and remains in character during the entire interaction. The initial prompt introduced two core interaction mechanisms: (1) sending the keyword "Hello" triggers the *Nature* to return a short textual prompt that provides cues; (2) sending the keyword "is that you", together with a photograph, prompts *Nature* to evaluate whether the submitted image matches *Nature* themselves. When responding to image submissions, *Nature* first evaluated the result. When no match is identified, *Nature* returned an extra cue prompt; When a match is identified, *Nature* provides a formal introduction describing the key characteristics of chanterelles. In actual use, users might interact using more natural language, and their messages are not restricted to the predefined keywords. All cue prompts were selected from a predefined prompt list, with repetition avoided. The noticing prompts included were of different styles, and included for example: "Can you find something golden yellow around you right now," and "You will find me where the forest fold gently" and "I'm very shy, small enough to hide the moss or fallen leaves". To enable a more seamless interaction and to simplify the development process of probe distribution, the prototype was implemented using a custom GPT built on ChatGPT.

5 Findings

In analyzing the test sessions, three main findings were discovered that will be discussed below. These findings then inform two design principles presented in the discussion. In this paper, the main focus will be on finding 1 and 2. As finding 3 highly affected the situation, it is still presented here, even though this will mostly be addressed in future work.

5.1 Finding 1: Stuck in Place

During the test session, a tendency for participants to become "stuck in place" was observed, in the sense that they circled within

a limited area rather than moving through the forest, which constrained their engagement with the wider forest environment. For most of the session, they remained relatively close to each other, and the researchers. Although researchers attempted to lead them forward by walking ahead, limited motivation to move forward was observed during their interaction with the system. The absence of a clear directional guiding role may further help explain this outcome. Neither participant was an experienced forager or hiker, and they had limited overall outdoor experience. In interviews, the participants mentioned their prior mushroom-picking experience, where they were accompanied by a more experienced forager who not only guided their movement through the forest but also confirmed whether collected mushrooms were edible. During the session, they appeared to expect the system to fulfill a similar guiding role. Another factor that may have strengthened this sense of lacking guidance relates to how the test began: the experience started directly off-trail, in the middle of the forest. Without the guidance provided by the forest's way-finding features, such as paths and trail markers, participants lacked orientation cues and quickly shifted their focus to small-scale details prompted by the system, thereby becoming uncertain about where to go next.

5.2 Finding 2: Attention Locked to the Ground

In terms of attention, it could be observed that participants' focus remained narrowly fixed on the ground, which reduced their connection with the surrounding nature environment. This was also mentioned in the interviews. Participant 1 described that he was able to notice more only after he gave up searching, at that moment when he became certain that he was unlikely to find anything during the session. [...] *I focused closely on the places it mentioned and the characteristics described. In some ways, that means I could overlook other things around me [...] I kind of gave up in the middle, and just roaming around, looking at the trees and all (other things).* (Participant 1, Post-Interview) Several observation may help describe this pattern. First, the probes provided by the system mostly focused on fine-grained details which require a high level of concentrated attention, examples are color, weight, scent, and size. The demand for attentional focus became even more pronounced under the snowy conditions. Second, as Participant 1 described based on his prior mushroom-picking experience in the interview, their attention during foraging was directed toward the ground, which was similar to their attention pattern during this session. In contrast, Participant 1 compared it with the other outdoor activities such as hiking, where attention is more often oriented toward the broader landscape. Additionally, many of the probes explicitly steered participants' attention toward ground-level details through phrases such as "hidden under moss or fallen leaves," "hidden in the shadow," and "let your gaze drift lower."

5.3 Finding 3: Snow Disrupted the Intended Experience

Unexpected early winter weather occurred the day before the test session, covering the entire forest area with about 10 cm of snow. The sudden shift to cold and snow resulted in changing the participants expectations on what to find. The snow covered ground

obscured most of the visual cues that were intended to guide participants' noticing, such as "moss" and "gold (color)". The probes that were intentionally designed to attune participants' embodied senses, such as scent and moisture, were largely muted by the cold weather. Therefore, the participants express low expectations in finding mushroom during the post-interview. This outcome was largely due to the seasonal mismatch between the design process and the test (and a bit of bad luck). Most of the researchers' exploratory work during the brainstorming phase took place in summer. Although a series of initial internal tests were conducted in the late autumn, the specific environmental conditions during winter were largely overlooked. As a result, the system was not fully adapted to the seasonal context in a snowy late-November, which is typically considered the very end of the chanterelles foraging season.

6 Discussion: Expanding the Understanding of Noticing in Nature

This section describes how reflections from this foraging-inspired noticing system can contribute to the broader area of more-than-human noticing. The system primarily focused on guiding participants' attention through detail-focused cues, which emphasized what we call *micro-level noticing* and, in turn, constrained movement and narrowed attention. This limitation was further amplified by the seasonal mismatch, which resulted in a different noticing experience of the surrounding nature environment than was expected. Similar forms of noticing can be seen in many previous studies [8, 11, 30, 34, 35]. While *micro-level noticing* can successfully help people attend to details that are often overlooked in everyday life, the present study suggests that to solely rely on micro-level focus may limit broader engagement with nature. "Where the forest folds" was one of the foraging prompts used in this study that captures what we learned through the design process. It refers to what we describe as macro-level noticing: the broader landscape conditions that support chanterelle growth and the layering between these features and small, detailed cues (such as color and shape) that guide foraging. We believe this complementary *macro-level noticing* is needed, which supports movement through space, and fluid shifts between scales of attention in order to encourage noticing more in the nature. Based on this perspective, two design principles are proposed.

6.1 Principle 1: Noticing More through Movement in Space

In nature environments, the breadth of the area available for exploration affords greater opportunities for discovery and encounter, which in turn can foster more noticing. Design should therefore encourage bodily and spatial movement to support exploration across the changing landscape. Such movement is encouraged by the affordances already present in the forest environment. These affordances operate in at least two ways. First, through the forest's way-finding features, such as paths and trail markings, which provide implicit directional cues for movements and orientation. Second, through existing outdoor activities where noticing can be layered on top, as is also shown in related work [11, 12, 25]. Therefore, our suggestion for noticing in nature is not solely defined by the activities themselves, but through providing an additional

mode of engagement with nature, while users navigate and explore through their original activity.

6.2 Principle 2: Noticing More through Fluidity Between Scale

The design should also encourage noticing through rhythmic shifts between macro- and micro-level perspective. As Finding 1 pointed out, the macro level involves perceiving the broader landscape that supports movements and orientation, whereas the micro level concerns attending to small, fine-grained details. Both are critical for encouraging noticing in the forest. Macro-level noticing supports breadth, enabling exploration across time and physical space, while micro-level noticing contributes to richness, offering more in-depth engagement with the nature environment. A disconnection between each scale can be seen in current design, while from a more experienced forager's perspective, foraging is often described as a continuous process of zooming in and out, in which attention fluidly shifts between micro-level details and a more macro sense of the natural environment. The typical process begins with observing broader landscape characteristics, such as terrain, moisture conditions, and co-living organisms that may support growth, and then gradually shifts toward a closer attention of potential foraging targets. Such noticing pattern should be accounted for when designing for noticing probes in the nature.

7 Conclusion

In this work, we ask *how the direct nature experience of foraging activities can inform more-than-human design and research, and more specifically, the method of the Art of Noticing*. Informed by a design probe tested in natural forest environments, this paper has presented two design principles for designing for expanding noticing in nature through: (1) noticing more through movement in space; (2) noticing more through fluidity between macro- and micro-level perspectives. These principles suggest that technologies for noticing in nature should not merely direct attention toward predefined, detailed targets, but rather support how attention unfolds while people move, explore, and shift perspective in natural environment.

Treating the re-design of the system as an example, building on these principles, first, the system could be layered onto other existing outdoor activities, where users are already moving and exploring the natural environment, thereby making better use of the affordances offered by the forest. In practice, this could mean designing the system as noticing probes that accompany activities such as hiking or trail running. Second, the system should support a rhythm of zooming in and out of the surroundings. This could be achieved by alternately providing noticing probes at different scales. For example, shifting between prompts that invite attention to small textures on the ground and those that address broader landscape features.

While this study set in Sweden, and grounded in a specific relationship to nature, with further research we believe these design principles can be more broadly adapted also to other cultural contexts when conducting more-than-human design directly in the nature.

References

- [1] Ecotype AB. 2025. Hitta Skog – Appen för att upptäcka nya skogar. <https://hittaskog.se/>. Accessed: 6 November 2025.
- [2] Heidi R. Biggs, Jeffrey Bardzell, and Shaowen Bardzell. 2021. Watching Myself Watching Birds: Abjection, Ecological Thinking, and Posthuman Design. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems* (Yokohama, Japan) (CHI '21). Association for Computing Machinery, New York, NY, USA, Article 619, 16 pages. doi:10.1145/3411764.3445329
- [3] Heidi R. Biggs, Anton Poikolainen Rosén, Emilija Veselova, David Sánchez Ruano, and Katerina Cerna. 2024. Multispecies Ethnography in Design Research and Practice. In *More-Than-Human Design in Practice* (1 ed.). Routledge, London, 119–131. doi:10.4324/9781003467731-12
- [4] Virginia Braun and Victoria Clarke. 2006. Using thematic analysis in psychology. *Qualitative research in psychology* 3, 2 (2006), 77–101.
- [5] Matthew H. E. M. Browning, Katherine J. Minnaugh, Carena J. van Riper, Heidemarie K. Laurent, and Steven M. LaValle. 2020. Can Simulated Nature Support Mental Health? Comparing Short, Single-Doses of 360-Degree Nature Videos in Virtual Reality With the Outdoors. *Frontiers in Psychology* Volume 10 - 2019 (2020). doi:10.3389/fpsyg.2019.02667
- [6] Christopher Frauenberger. 2019. Entanglement HCI The Next Wave? *ACM Trans. Comput.-Hum. Interact.* 27, 1, Article 2 (Nov. 2019), 27 pages. doi:10.1145/3364998
- [7] Terry Hartig, Richard Mitchell, Sjørp de Vries, and Howard Frumkin. 2014. Nature and Health. *Annual Review of Public Health* 35, Volume 35, 2014 (2014), 207–228. doi:10.1146/annurev-publhealth-032013-182443
- [8] Youyang Hu, Chiao-chi Chou, and Yasuaki Kakehi. 2023. Synplant: Cymatics Visualization of Plant-Environment Interaction Based on Plants Biosignals. *Proc. ACM Comput. Graph. Interact. Tech.* 6, 2, Article 26 (Aug. 2023), 7 pages. doi:10.1145/3597622
- [9] Karin Johansson and Jon Back. 2024. Daddy, You Can Be the Fox: Designing Hybrid Playgrounds for Nature Integrated Storytelling. In *Proceedings of the 13th Nordic Conference on Human-Computer Interaction* (Uppsala, Sweden) (NordiCHI '24). Association for Computing Machinery, New York, NY, USA, Article 27, 17 pages. doi:10.1145/3679318.3685360
- [10] J. F. Kelley. 1983. An empirical methodology for writing user-friendly natural language computer applications. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Boston, Massachusetts, USA) (CHI '83). Association for Computing Machinery, New York, NY, USA, 193–196. doi:10.1145/800045.801609
- [11] Jen Liu, Daragh Byrne, and Laura Devendorf. 2018. Design for Collaborative Survival: An Inquiry into Human-Fungi Relationships. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems* (Montreal QC, Canada) (CHI '18). Association for Computing Machinery, New York, NY, USA, 1–13. doi:10.1145/3173574.3173614
- [12] Szu-Yu (Cyn) Liu, Shaowen Bardzell, and Jeffrey Bardzell. 2019. Symbiotic Encounters: HCI and Sustainable Agriculture. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems* (Glasgow, Scotland UK) (CHI '19). Association for Computing Machinery, New York, NY, USA, 1–13. doi:10.1145/3290605.3300547
- [13] Angella Mackey, David Ng McCallum, Oscar Tomico, and Martijn De Waal. 2025. What Comes After Noticing?: Reflections on Noticing Solar Energy and What Came Next. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. ACM, Yokohama Japan, 1–14. doi:10.1145/3706598.3713239
- [14] Elizabeth McAllister, Navjot Bhullar, and Nicola S. Schutte. 2017. Into the Woods or a Stroll in the Park: How Virtual Contact with Nature Impacts Positive and Negative Affect. *International Journal of Environmental Research and Public Health* 14, 7 (2017). doi:10.3390/ijerph14070786
- [15] Johanna Nicenboim, Doenja Oogjes, Heidi Biggs, and Seowoo Nam. 2025. Decentering Through Design: Bridging Posthuman Theory with More-than-Human Design Practices. *Human-Computer Interaction* 40, 1-4 (July 2025), 195–220. doi:10.1080/07370024.2023.2283535
- [16] Claudia Núñez Pacheco and Anton Poikolainen Rosén. 2024. Articulating Felt Senses for More-Than-Human Design: A Viewpoint for Noticing. In *Proceedings of the 2024 ACM Designing Interactive Systems Conference* (Copenhagen, Denmark) (DIS '24). Association for Computing Machinery, New York, NY, USA, 1029–1043. doi:10.1145/3643834.3661554
- [17] Claudia Núñez-Pacheco. 2025. Entangled Weathers: A Noticing Tactic. In *Proceedings of the 2025 Conference on Creativity and Cognition*. ACM, Virtual United Kingdom, 694–707. doi:10.1145/3698061.3726955
- [18] Kenton O'Hara. 2008. Understanding geocaching practices and motivations. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Florence, Italy) (CHI '08). Association for Computing Machinery, New York, NY, USA, 1177–1186. doi:10.1145/1357054.1357239
- [19] Anton Poikolainen Rosén, Antti Salovaara, Andrea Botero, and Marie Louise Juul Søndergaard. 2024. Introduction. In *More-Than-Human Design in Practice* (1 ed.). Routledge, London, xix–xxxi. doi:10.4324/9781003467731-1
- [20] Elena Sabinson, Isha Pradhan, and Keith Evan Green. 2021. Plant-Human Embodied Biofeedback (pheB): A Soft Robotic Surface for Emotion Regulation in Confined Physical Space. In *Proceedings of the Fifteenth International Conference on Tangible, Embedded, and Embodied Interaction* (Salzburg, Austria) (TEI '21). Association for Computing Machinery, New York, NY, USA, Article 89, 14 pages. doi:10.1145/3430524.3446065
- [21] Sigurjónsdóttir, Dagný Bryndis. 2021. The Swedish Allemansrätt: An Embodiment of Swedishness. Student Paper.
- [22] Statens offentliga utredningar. 1940. SOU 1940:12 — 268. Swedish Government Official Reports. Report on Allemansrätten (Public Access Rights).
- [23] Anna Sténs and Camilla Sandström. 2014. Allemansrätten in Sweden: A Resistant Custom. *Landscapes* 15, 2 (2014), 106–118. doi:10.1179/1466203514Z.00000000029 arXiv:https://doi.org/10.1179/1466203514Z.00000000029
- [24] Alex Taylor. 2015. After interaction. *Interactions* 22, 5 (Aug. 2015), 48–53. doi:10.1145/2809888
- [25] Anna Lowenhaupt Tsing. 2015. *The mushroom at the end of the world: on the possibility of life in capitalist ruins*. Princeton University Press, Princeton.
- [26] Annika Waern, Fredrika Mårtensson, Jon Back, Anna Litsmark, and Eva-Lotta Sallnäs Pysander. 2025. Digital Play in Nature: A Study of Digital Play Installations from a Nature Play Perspective. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems* (CHI '25). Association for Computing Machinery, New York, NY, USA, Article 324, 19 pages. doi:10.1145/3706598.3713303
- [27] Ron Wakkary, Doenja Oogjes, Nazmus Sakib, and Armi Behzad. 2023. Turner Boxes and Bees: From Ambivalence to Diffraction. In *Proceedings of the 2023 ACM Designing Interactive Systems Conference* (Pittsburgh, PA, USA) (DIS '23). Association for Computing Machinery, New York, NY, USA, 790–807. doi:10.1145/3563657.3596081
- [28] Ron Wakkary, Doenja Oogjes, Oscar Tomico, Nazmus Sakib, and Ege Kökel. 2025. Backyard Practices: A Liminal Approach to Designing in More-than-Human Worlds. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems* (CHI '25). Association for Computing Machinery, New York, NY, USA, Article 768, 18 pages. doi:10.1145/3706598.3713291
- [29] Sarah Webber, Ryan M. Kelly, Greg Wadley, and Wally Smith. 2023. Engaging with Nature through Technology: A Scoping Review of HCI Research. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany) (CHI '23). Association for Computing Machinery, New York, NY, USA, Article 521, 18 pages. doi:10.1145/3544548.3581534
- [30] Xinquan Wen and Yiyang Wu. 2022. Slowing Plants, Slowing Home: Proposing A Plant-Decentred Perspective to Human-Plant Interaction. In *Adjunct Proceedings of the 2022 Nordic Human-Computer Interaction Conference* (Aarhus, Denmark) (NordiCHI '22 Adjunct). Association for Computing Machinery, New York, NY, USA, Article 47, 5 pages. doi:10.1145/3547522.3547691
- [31] Ying Yang, Huajian Cai, Ziyang Yang, Xiaochong Zhao, Mei Li, Rui Han, and Sylvia Xiaohua Chen. 2022. Why does nature enhance psychological well-being? A Self-Determination account. *Journal of Environmental Psychology* 83 (2022), 101872. doi:10.1016/j.jenvp.2022.101872
- [32] Yangyang Yang and Kimiko Ryokai. 2025. Being The Creek: Mobile Augmented Reality Experience as an Invitation for Exploring More-Than-Human Perspectives. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems* (CHI '25). Association for Computing Machinery, New York, NY, USA, Article 740, 19 pages. doi:10.1145/3706598.3713713
- [33] EDENAPP Sp. z o.o. 2025. Eden – Edible Plants, Healthy Products, and Web3 (App). <https://oureden.io/app>. Accessed: 6 November 2025.
- [34] Jiwei Zhou, Zjenja Doubrovski, Elisa Giaccardi, and Elvin Karana. 2024. Living with Cyanobacteria: Exploring Materiality in Caring for Microbes in Everyday Life. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '24). Association for Computing Machinery, New York, NY, USA, Article 561, 20 pages. doi:10.1145/3613904.3642039
- [35] Jingwen Zhu, Samantha Chang, Ruth Zhao, and Cindy Hsin-Liu Kao. 2025. LivingLoom: Investigating Human-Plant Symbiosis Through Integrating Living Plants Into (E-)Textiles. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems* (CHI '25). Association for Computing Machinery, New York, NY, USA, Article 773, 18 pages. doi:10.1145/3706598.3713156
- [36] Heather Toomey Zimmerman, Susan M. Land, Michael R. Mohney, Gi Woong Choi, Chrystal Maggiore, Soo Hyeon Kim, Yong Ju Jung, and Jaclyn Dudek. 2015. Using augmented reality to support observations about trees during summer camp. In *Proceedings of the 14th International Conference on Interaction Design and Children* (Boston, Massachusetts) (IDC '15). Association for Computing Machinery, New York, NY, USA, 395–398. doi:10.1145/2771839.2771925